

* Spherical solutions: Reduce symmetry so that all we require is (r, t) in 4D. For $D \geq 4$ the remaining variables are angles on S^{D-2}

S^n geometry

(i) There are 1-forms $\bar{e}^a$, $a=1, \dots, n$ s.t.

$$d\Omega_n^2 = \delta_{ab} \bar{e}^a \bar{e}^b$$

(ii) Torsion free connection 1-form $\bar{\omega}^a_b$ which satisfy the Cartan structure eq.

$$d\bar{e}^a + \bar{\omega}^a_b \wedge \bar{e}^b = 0$$

(iii) S^n is maximally symmetric so its curvature 2-form is

$$\bar{\rho}^{ab} = \bar{e}^a \wedge \bar{e}^b$$

* $R_{\mu\nu\alpha\beta} = \partial_\mu \omega_{\nu\alpha\beta} - \partial_\nu \omega_{\mu\alpha\beta} + \omega_{\mu\alpha\gamma} \omega_{\nu\beta}^\gamma - \omega_{\nu\alpha\gamma} \omega_{\mu\beta}^\gamma$
and so

$$\bar{\rho}^{ab} \equiv \frac{1}{2} (R_{\mu\nu})^{ab}(x) dx^\mu \wedge dx^\nu$$

and

$$d\bar{\omega}^{ab} + \bar{\omega}^a_c \wedge \bar{\omega}^{cb} = \bar{\rho}^{ab}$$

$$\bar{\rho}^{ab} \wedge \bar{e}_b = 0$$

* A maximally symmetric spacetime admits vector fields $\mathcal{L}_X(g)_{\mu\nu} = 0$

$\frac{n(n+1)}{2}$ Killing

* Minkowski case: (n) - translations $\binom{n}{2}$: boosts
 $\rightarrow n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$

* Means that $R_{\mu\nu\rho\sigma} = K(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$

* $g_{\mu\nu} = e_\mu^a e_\nu^b \delta_{ab}$

* $\dim(\text{Solution}) = \frac{n(n+1)}{2}$

ex: S^2 : lcs embedding coordinates $(x^1)^2 + (x^2)^2 + (x^3)^2 = 1$

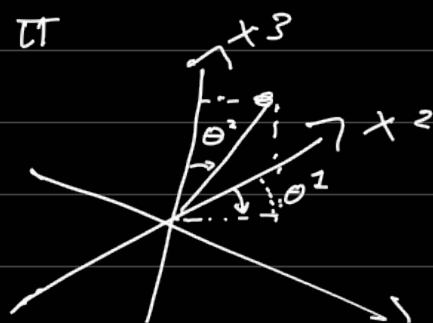
θ^1, θ^2 : $0 \leq \theta^1 \leq 2\pi$, $0 \leq \theta^2 \leq \pi$

$(x^1 = y, x^2 = x, x^3 = z)$

$x^1 = \sin(\theta^2) \sin(\theta^1)$

$x^2 = \sin(\theta^2) \cos(\theta^1)$

$x^3 = \cos(\theta^2)$



$$d\Omega_2^2 = dx_1^2 + dx_2^2 + dx_3^2$$

$$= (d\theta^2)^2 + \sin^2(\theta^2)(d\theta^1)^2$$

• define frame 1-forms

$$\bar{e}^2 = d\theta^2$$

$$\bar{e}^1 = \sin(\theta^2) d\theta^1$$

$$\left| \begin{array}{l} e^a \equiv e^a_\mu dx^\mu \\ E_a = e^{\mu}_a \frac{\partial}{\partial x^\mu} \end{array} \right| \langle e^a | E_b \rangle = \delta^a_b$$

then

$$d\Omega_2^2 = \bar{e}^2 \otimes \bar{e}^2 + \bar{e}^1 \otimes \bar{e}^1$$

$$\text{and } \bar{e}^1_{\theta^1} = \sin(\theta^2) \quad \bar{e}^1_{\theta^2} = 0$$

$$\bar{e}^2_{\theta^1} = 0$$

$$\bar{e}^2_{\theta^2} = 1$$

$$\bar{e}^a_\mu = \begin{pmatrix} \sin(\theta^2) & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{From } d\bar{e}^a + \omega^a_b \wedge \bar{e}^b = 0$$

$$d\bar{e}^a = \cos(\theta^2) d\theta^1 \wedge d\theta^2 = d\bar{e}^1$$

$$\begin{aligned} \omega^{a b} \wedge e_b &= \omega^{a 2} \wedge e_1 + \omega^{a 2} \wedge e_2 \\ &= \omega^{a 2} \wedge (\sin(\theta^2) d\theta^1 + d\theta^2) \\ &= \omega^{1 2} \wedge d\theta^2 \end{aligned}$$

$$\text{and so } \bar{\omega}^{1 2} = \cos(\theta^2) d\theta^1$$

* For S^n : $\sum_{i=1}^{n+1} (x_{(n)}^i)^2 = 1 \Leftrightarrow \mathbb{R}^{n+1} \quad x_{(n)}^i$

• we recursively define them via lower dim spheres.
and principal angle θ^n , $x_{(n-1)}^i \in S^{n-1}$

$$x_{(n)}^{n+1} = \cos(\theta^n), \quad x_{(n)}^a = \sin(\theta^n) x_{(n-1)}^a, \quad a \leq n, \quad 0 \leq \theta^n \leq \pi$$

• This exactly defines coordinates θ^a , $a = 1, \dots, n$ which are called
hyperspherical coordinates

$$0 \leq \theta^1 < 2\pi \quad \text{and} \quad 0 \leq \theta^a \leq \pi \quad \forall a \geq 1.$$

$$d\Omega_n^2 = (d\theta^n)^2 + \sin^2(\theta^n) d\Omega_{n-1}^2$$

$$\bar{e}_{(n)}^n = d\theta^n$$

$$\bar{e}_{(n)}^a = \sin(\theta^n) e_{(n-1)}^a$$

$$a \leq n-1$$

$$\bar{\omega}_{(n)}^{ab} = \bar{\omega}_{(n-1)}^{ab}$$

$$\omega_{(n)}^{an} = \cos(\theta^n) \bar{e}_{(n-1)}^a$$

$$S^2: \bar{e}_{(2)}^2 = d\theta^2, \bar{e}_{(2)}^1 = \sin(\theta^2) d\theta^1$$

$$\bar{\omega}_{(2)}^{a2} = \cos(\theta^2) \bar{e}_{(2)}^a \rightarrow \bar{\omega}_{(2)}^{12} = \cos(\theta^2) d\theta^1$$

$$S^3: \bar{e}_{(3)}^3 = d\theta^3, \bar{e}_{(3)}^2 = \sin(\theta^3) d\theta^2$$

$$\bar{e}_{(3)}^2 = \sin(\theta^3) \bar{e}_{(2)}^1 = \sin(\theta^3) \sin(\theta^2) d\theta^1$$

$$\bar{\omega}_{(3)}^{03} = \cos(\theta^3) \bar{e}_{(2)}^a$$

$$\begin{cases} \bar{\omega}_{(3)}^{13} = \cos(\theta^3) \sin(\theta^2) d\theta^1 \\ \bar{\omega}_{(3)}^{23} = \cos(\theta^3) d\theta^2 \end{cases}$$

$$\omega_{(3)}^{12} = \omega_{(2)}^{12} = \cos(\theta^2) d\theta^1$$

$$e_\mu^a = \begin{pmatrix} \sin(\theta^3) \sin(\theta^2) & & \\ & \sin(\theta^3) & \\ & & 1 \end{pmatrix}$$

* Compactly:

$$\bar{e}^a = \left(\prod_{j=a+1}^n \sin(\theta^j) \right) d\theta^a \quad a \leq n$$

$$\bar{\omega}^{ab} = \cos(\theta^b) \left(\prod_{j=a+1}^{b-1} \sin(\theta^j) \right) d\theta^a \quad 1 \leq a < b \leq n$$

ex: $S^6: n=6$

$$\bar{e}^a = \sin(\theta^6) \sin(\theta^5) \dots \sin(\theta^{a+1}) d\theta^a$$

and calculate $\rho^{ab} = \frac{1}{2} R_{\mu\nu}{}^{ab} dx^\mu \wedge dx^\nu$ and compare to

$$d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb} = \rho^{ab}$$

Claim: The volume of S^n is

$$\Omega_n \equiv \int \prod_{a=1}^n d\theta^a \sqrt{g} = \int \bar{e}^1 \wedge \dots \wedge \bar{e}^n = \frac{2\pi^{(n+1)/2}}{\Gamma((n+1)/2)}$$

proof:

$$d\Omega_n^2 = \delta_{ab} \bar{e}^a \bar{e}^b = g_{\mu\nu} dx^\mu dx^\nu$$

$$dV = \bar{e}^0 \wedge \bar{e}^1 \wedge \dots \wedge \bar{e}^{n-1}$$

$$= \delta_{a_1 \dots a_{n-1}}^{0 \dots n-1} e^{a_0} \wedge \dots \wedge e^{a_{n-1}}$$

$$= \frac{1}{(n-1)!} \varepsilon_{a_1 \dots a_{n-1}} e^{a_0} \wedge \dots \wedge e^{a_{n-1}}$$

$$= \frac{1}{(n-1)!} \varepsilon_{a_1 \dots a_{n-1}} e^{a_0} \wedge \dots \wedge e^{a_{n-1}} (dx^{\mu_0} \wedge \dots \wedge dx^{\mu_{n-1}})$$

$$= \frac{1}{(n-1)!} \varepsilon_{\mu_0 \dots \mu_{n-1}} dx^{\mu_0} \wedge \dots \wedge dx^{\mu_{n-1}}$$

$$= e dx^0 \wedge \dots \wedge dx^{n-1}$$

$$= d^p x \sqrt{-g}$$

$$e = \sqrt{-g} = \det(e_\mu^a)$$

Here $dx^0 = d\theta^0$ etc...

* Now need to expand in θ 's to get a formula to get the volume

$$e = \sin(\theta^n)^{n-1} \sin(\theta^{n-1})^{n-2} \dots \sin(\theta^2)^1$$

$$\rightarrow (\pi^{1/2})^{n-1} / (\pi^{1/2})^{n-2} \dots (\pi^{1/2})^1$$

AdS_D

- Spherically symmetric metric: solutions to E.o.M to the gravitational action

$$S = \frac{1}{2\kappa^2} \int d^D x \sqrt{-g} (R - \Lambda)$$

$$\Lambda = -(D-1)(D-2)/L^2$$

$$\begin{cases} \Lambda < 0 & \text{in AdS} \\ \Lambda = 0 & \text{flat } \mathbb{E} \\ \Lambda > 0 & \text{dS} \end{cases}$$

$$\delta S = \frac{1}{2\kappa^2} \int d^D x \delta(\sqrt{-g}) (R - \Lambda) + \frac{1}{2\kappa^2} \int d^D x \sqrt{-g} (\delta R)$$

$$\bullet \delta R = \delta g^{\mu\nu} R_{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}$$

$$\delta R_{\mu\nu} = \nabla_\rho \delta \Gamma_{\mu\nu}^\rho - \nabla_\mu \delta \Gamma_{\nu\rho}^\rho$$

→ Vanishes on boundary since $\nabla_\alpha g = 0$

$$\delta \sqrt{-g} = -\frac{1}{2} \frac{1}{\sqrt{-g}} \delta \det(g) = -\frac{1}{2} \frac{1}{\sqrt{-g}} \delta e^{\text{tr}(R(g))}$$

$$= -\frac{1}{2} \frac{1}{\sqrt{-g}} (\text{tr}(g)) \text{Tr}(g^{-1} \delta g)$$

$$= -\frac{1}{2} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$$

$$\star \int d^D x \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} (R - \Lambda) + \sqrt{-g} R_{\mu\nu} \right) \delta g^{\mu\nu}$$

$$\Rightarrow R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} (R - \Lambda) = 0$$

$$R - \frac{1}{2} D(R) = -\frac{1}{2} D\Lambda$$

$$2R - DR = D\Lambda \Rightarrow R = \frac{D\Lambda}{D-2}$$

$$R_{\mu\nu} = \frac{1}{2} g_{\mu\nu} \left(\frac{D - D + 2}{D-2} \right) \Lambda$$

$$R_{\mu\nu} = g_{\mu\nu} \frac{\Lambda}{D-2}$$

$$\text{with } \Lambda = -\frac{(D-1)(D-2)}{L^2}$$

we have

$$R_{\mu\nu} = -\frac{(D-1)}{L^2} g_{\mu\nu}$$

- So AdS is maximally symmetric with negative curvature:

$$\begin{aligned} R_{\mu\nu\rho\sigma} &= K (g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}) \\ \Rightarrow R_{\mu\nu} &= K(D-1) g_{\mu\nu} \\ R &= K(D-1) D \end{aligned}$$

$$K = -\frac{1}{L^2}$$

AdS_D obtained from its Embedding in $\mathbb{R}^{D+1}$:

$$\eta_{AB} = \text{diag}(-+ \dots + -)$$

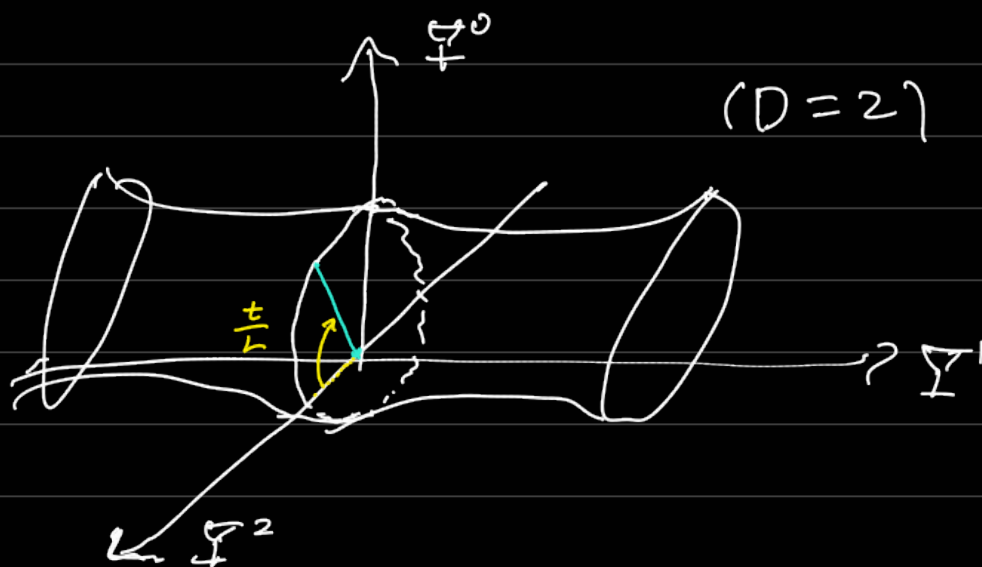
$$: \text{SO}(D-1, 2)$$

$$A, B = 0, \dots, D$$

• Cartesian coordinates $\mathbb{I}^A$

• The hyperboloid which defines the AdS_D is the Surface:

$$f(\mathbb{I}) = \mathbb{I}^A \eta_{AB} \mathbb{I}^B = -(\mathbb{I}^0)^2 + \sum_{i=1}^{D-1} (\mathbb{I}^i)^2 - (\mathbb{I}^D)^2 = -L^2$$



• Invariant under $\mathbb{I}'^A = \Lambda^A_B \mathbb{I}^B$

$$\mathbb{I}'^A \eta_{AB} \mathbb{I}'^B = \Lambda^A_C \mathbb{I}^C \eta_{AB} \Lambda^B_D \mathbb{I}^D$$

$$\Rightarrow \Lambda^A_C \eta_{AB} \Lambda^B_D = \eta_{CD} \quad \text{so} \quad \Lambda^A_B \in \text{SO}(D-1, 2)$$

$$\Lambda^T \eta \Lambda = \eta$$

• Isometry group of AdS_D is $\text{SO}(D-1, 2)$

$$\mapsto \mathbb{I}'^A = (e^{i\omega_{AB} L^{AB}}) \mathbb{I}^A$$

We obtain the same generators for this group

$$L_{AB} = \mathbb{I}_A \frac{\partial}{\partial \mathbb{I}^B} - \mathbb{I}_B \frac{\partial}{\partial \mathbb{I}^A}$$

• Note $L_{AB} f(\mathbb{I}) = 0$

Proof:

$$\begin{aligned} L_{AB} (\mathbb{I}^C \eta_{CD} \mathbb{I}^D) &= \mathbb{I}_A (2 \eta_{BC} \mathbb{I}^D) - \mathbb{I}_B (2 \eta_{AC} \mathbb{I}^D) \\ &= 2 (\mathbb{I}_A \mathbb{I}_B - \mathbb{I}_B \mathbb{I}_A) = 0 \end{aligned}$$

• AdS is homogeneous: any two points can be related by an isometry G

• isotropy group: (little?) $H \subset G$ leaves a point

invariant

→ The manifold can be identified as G/H

• Maximal compact subgroup of $SO(D-1, 2)$ is

$$\underbrace{SO(D-1)}_{\text{rotates } \gamma^i} \times \underbrace{SO(2)}_{\text{rotates } 0-D \text{ plane}} \subset SO(D-1, 2)$$

(rotates γ^i
 $i=1, \dots, D-1$)
(little group)

(Rotates 0-D plane)

* An arbitrary point $\mathbb{I}^A$ can be brought to $(L, 0, \dots, 0)$

1) $\mathbb{I}^A \rightarrow (\gamma'^0, \gamma'^1, 0, \dots, 0)$ with $\gamma'^0 > 0$

$$\begin{pmatrix} \gamma^0 \\ \vdots \\ \gamma^D \end{pmatrix} = \begin{pmatrix} \gamma'^0 \\ \gamma'^1 \\ \vdots \end{pmatrix} \rightarrow \begin{pmatrix} L \\ 0 \\ \vdots \end{pmatrix}$$

$$\begin{pmatrix} A & D \\ B & D \end{pmatrix} \begin{pmatrix} \gamma'^0 \\ \gamma'^1 \end{pmatrix} = \begin{pmatrix} L \\ 0 \end{pmatrix}, \quad A = \gamma'^0 / L, \quad B = \gamma'^1 / L$$

* Note that we required $\gamma'^0 > 0$, so for $\gamma'^D > 0$ we could also have $(0, \dots, L)$

* Intrinsic coordinates x^μ , $\mu = 0, \dots, D-1$ on hyperboloid
 L_{AB} expressed via $x^\mu \rightarrow L_{\mu\nu}$ are Killing vect. fields

(Similar to $\gamma^1 = R \cos(\theta)$ on S^2 etc...)

$\gamma^i = r \bar{x}^i$ with	$\sum_{i=1}^{D-1} (\bar{x}^i)^2 = 1$
$\gamma^0 = \sqrt{L^2 + r^2} \sin\left(\frac{t}{L}\right)$	$\gamma^D = \sqrt{L^2 + r^2} \cos\left(\frac{t}{L}\right)$

• $\bar{x}^i$ parametrize S^{D-2} : $\bar{x}_{(D-1)}^{D-2} = \cos(\theta^{D-1})$
 $\bar{x}_{(D-1)}^u = \sin(\theta^{D-1}) \bar{x}_{(D-2)}^u$

* Induced metric: $dY^A \eta_{AB} dY^B$
 $-d\gamma^0{}^2 + r^2 d\Omega_{D-1}^2 - d\gamma^D{}^2$

• $d\gamma^0 = \frac{1}{\sqrt{r}} r \sin\left(\frac{t}{L}\right) + \sqrt{\frac{1}{L^2 + r^2}} \cos\left(\frac{t}{L}\right)$

$$(d\gamma^0)^2 = \frac{r^2}{(L^2 + r^2)} \sin^2 \left(\frac{2r}{L} \cos \sin + \frac{L^2 + r^2}{L^2} \cos^2 \right) (dr)^2 + (dt)^2$$

$$(dy^D)^2 = \frac{r^2}{L^2 + r^2} \cos^2\left(\frac{t}{L}\right) - \frac{2r}{L} \sin\cos + \frac{L^2 + r^2}{L^2} \sin^2$$

$$(dr)^2 \quad dr dt \quad dt^2$$

$$\Rightarrow -\left(\frac{r^2}{L^2 + r^2}\right) dr^2 - \frac{L^2 + r^2}{L^2} dt^2$$

* Note $d\Omega_{D-1}^2 = dr^2 + r^2 \underbrace{d\Omega_{D-2}^2}_{\sum_{i=1}^{D-1} (d\bar{x}^i)^2}$

$$\Rightarrow dr^2 \left(1 - \frac{r^2}{L^2 + r^2}\right) = \frac{L^2}{L^2 + r^2} dr^2 = \left(1 + \frac{r^2}{L^2}\right)^{-1} dr^2$$

$$ds^2 = -\left(1 + \frac{r^2}{L^2}\right) dt^2 + \left(1 + \frac{r^2}{L^2}\right)^{-1} dr^2 + r^2 d\Omega_{D-2}^2$$

AdS embedding (induced) metric
 $\gamma_A \rightarrow (t, r, \bar{x}^i)$, $0 < r < \infty$
 $0 \leq t < 2\pi L$

• Contains timelike curves: fix r and $\bar{x}^i$ and go through t .

→ Unwrap the circle and pass to the covering space in which $-\infty < t < \infty$

* There is no origin, $r=0$ is not unique

$\underline{r=0} \equiv$
 $\gamma^0 = L \sin\left(\frac{t}{L}\right)$, $\gamma^D = L \cos\left(\frac{t}{L}\right)$
 we get a circle and $ds^2 = -dt^2$

• $t \rightarrow t + a$ is an isometry which generates Killing vector $\frac{\partial}{\partial t}$

$$t \rightarrow t + a = \left(1 + a \frac{\partial}{\partial t}\right) t$$

$$\mathcal{L}_{\frac{\partial}{\partial t}} g = 0 \quad ; \quad T = a \frac{\partial}{\partial t}$$

$$\Rightarrow L_{AB} = \gamma_A \frac{\partial}{\partial \gamma_B} - \gamma_B \frac{\partial}{\partial \gamma_A}$$

$$L_{0D} = \gamma_0 \frac{\partial}{\partial \gamma_D} - \gamma_D \frac{\partial}{\partial \gamma_0}$$

$$(\gamma_0)^2 + (\gamma^D)^2 = (L^2 + r^2)$$

$$r^2 = (y^0)^2 - (y^D)^2 - L^2$$

$$\frac{y^0}{y^D} = \tan\left(\frac{\pi}{2}\right)$$

• Then $\frac{1}{L} \frac{dy}{dt} = \frac{2}{L^2}$

$$\cosh^2 - \sinh^2 = 1$$

* Can admit different coordinate reps:

$$(i) \quad r \rightarrow y: \quad \cosh\left(\frac{y}{L}\right) = \sqrt{1 + \frac{r^2}{L^2}} \quad \left| \quad \sinh^2\left(\frac{y}{L}\right) = \frac{r^2}{L^2} \right. \\ \sinh\left(\frac{y}{L}\right) \frac{dy}{L} = \frac{1}{L^2} \frac{r^2}{\sqrt{1 + \frac{r^2}{L^2}}} dr \\ \sinh^2\left(\frac{y}{L}\right) dy^2 = (L^2 + r^2)^{-1} dr^2 \quad (0 \leq y < \infty) \\ dy^2 = \left(\frac{L^2}{L^2 + r^2} \right) dr^2 = \left(1 + \frac{r^2}{L^2} \right)^{-1} dr^2$$

$$* \quad dy^2 = \left(1 + \frac{r^2}{L^2} \right)^{-1} dr^2, \quad r = L \sinh\left(\frac{y}{L}\right)$$

$$ds^2 = -\cosh^2\left(\frac{y}{L}\right) dt^2 + dy^2 + L^2 \sinh^2\left(\frac{y}{L}\right) d\Omega_{D-2}^2$$

$$(ii) \quad y \rightarrow \rho: \quad \cosh\left(\frac{y}{L}\right) = \frac{1}{\cos(\rho)} \quad 0 \leq \rho < \pi/2 \\ \frac{dy}{L} \sinh\left(\frac{y}{L}\right) = \frac{1}{\cos^2(\rho)} \sin(\rho) d\rho \Rightarrow dy^2 \sinh^2\left(\frac{y}{L}\right) = \frac{L^2}{\cos^2(\rho)} \sin^2(\rho) d\rho^2 \\ \frac{\sin^2(\rho)}{\cos^2(\rho)} = \tanh^2(\rho) \\ \Rightarrow dy^2 = \frac{L^2}{\cos^2(\rho)} d\rho^2$$

• define $t = L\tau$

$$\cosh^2\left(\frac{y}{L}\right) dt^2 = \frac{L^2}{\cos^2(\rho)} d\tau^2$$

$$\Rightarrow ds^2 = \frac{L^2}{\cos^2(\rho)} \left(-d\tau^2 + (d\rho^2 + \sin^2(\rho) d\Omega_{D-2}^2) \right)$$

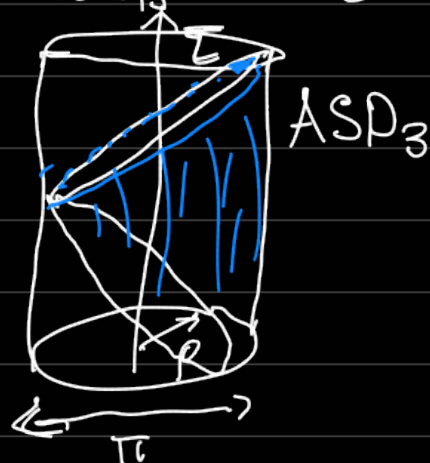
• Conformal factor: $L^2 / \cos^2 \rho = w$

* This metric is w x direct product of real line $-\infty < \tau < \infty$ x Sphere S^{D-1}

• This product spacetime is called the Einstein static Universe.

• $0 \leq \rho \leq \pi$ is needed to cover the full S^{D-1} , so the AdS metric includes only the upper hemisphere $\text{AdS} \approx \text{half of Einstein static universe}$

- Globally AdS_D is the interior of the cylinder



ex: $D=3$: $d\Omega_1^2 = d\theta^2$, $0 \leq \theta \leq \pi$

$$ds^2 = \frac{L^2}{\cos^2(\rho)} (-d\tau^2 + (d\rho^2 + \sin^2(\rho) d\theta^2))$$

- Each interior point represent an S^{D-3} (in $D>3$)
 $\rho_0 = \pi/2$: where ω is singular, so any interval length of any space-like curve becomes infinite as the curve approaches the boundary

$\Rightarrow$ keep (t, θ) fixed and vary ρ :

$$ds^2 = \frac{L^2}{\cos^2(\rho)} d\rho^2$$

as $\rho \rightarrow \pi/2$ $ds^2 \rightarrow \infty$

- Boundary is at an infinite spatial distance from any interior point

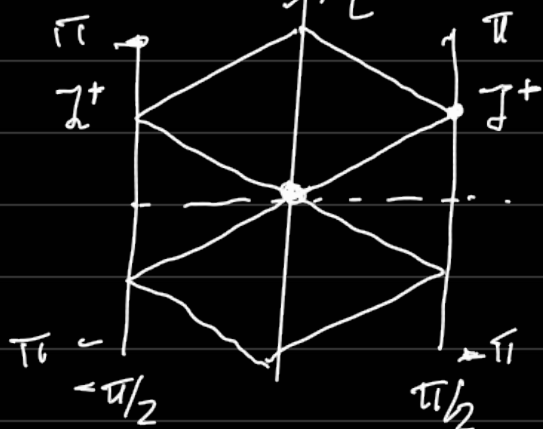
- Weyl rescale: $ds'^2 = L^2 d\rho^2$
 simply leaving $\mathbb{R} \times S^{D-2}$

- Constant angle Θ : Null geodesics amount to
 $d\tau = \pm d\rho$ $\tau = \pm \rho + c$



light rays prop at 45°

$\rho = \pi/2$: Spatial infinity



- Light reaches spatial ∞ in finite time $\tau = \pi/2$
- Information from $\mathcal{I}^+$ reaches $\circ$ in finite time
 $\Rightarrow$ Cauchy problem in AdS is ill defined.
- Need additional Boundary cond.

Poincaré Patch = Region where AdS is conformal to half of flat MD

$$Y^0 = L u x^0$$

$$Y^i = L u x^i, \quad i = 1, \dots, D-2$$

$$Y^{D-1} = \frac{1}{2u} (-1 + u^2 (L^2 - x^2))$$

$$Y^D = \frac{1}{2u} (1 + u^2 (L^2 + x^2))$$

$$x^2 = -(x^0)^2 + \sum_{i=1}^{D-1} (x^i)^2 \quad \leftarrow (D-1)\text{-Minkowski}$$

Time: $x^0 = t \in (-\infty, \infty)$

Space: $x^i \in (-\infty, \infty)$

$$0 < u < \infty$$

$\Rightarrow$ Restriction ^{of u} is needed to maintain a single valuedness of the hyperboloid Y^D and Y^i

$$-L^2 u^2 (x^0)^2 + \sum_{i=1}^{D-2} L^2 u^2 (x^i)^2 + \left(\frac{1}{2u} (-1 + u^2 (L^2 - x^2)) \right)^2 - \left(\frac{1}{2u} (1 + u^2 (L^2 + x^2)) \right)^2 =$$

$$= L^2 u^2 x^2 + \frac{1}{4u^2} (1 - 2u^2 (L^2 - x^2) + u^4 (L^2 - x^2)^2) - \frac{1}{4u^2} (1 + 2u^2 (L^2 + x^2) + u^4 (L^2 + x^2)^2)$$

$$= L^2 u^2 x^2 - \frac{1}{2} ((L^2 - x^2) + (L^2 + x^2)) + \frac{1}{4} u^2 (L^2 - x^2)^2 - \frac{1}{4} u^2 (L^2 + x^2)^2$$

$$= L^2 u^2 x^2 - L^2 + \frac{1}{4} u^2 ((L^2 - x^2)^2 - (L^2 + x^2)^2)$$

$$= L^2 u^2 x^2 - L^2 + \frac{1}{4} u^2 (-2L^2 x^2 - 2L^2 x^2)$$

$$= L^2 u^2 x^2 - L^2 - u^2 L^2 x^2 = -L^2 \quad \checkmark$$

* Poincaré metric: $ds^2 = L^2 \left[\frac{du^2}{u^2} + u^2 (-dx^0)^2 + \sum_{i=1}^{D-2} (dx^i)^2 \right]$

• define $z = 1/u$ $dz = -\frac{1}{u^2} du$

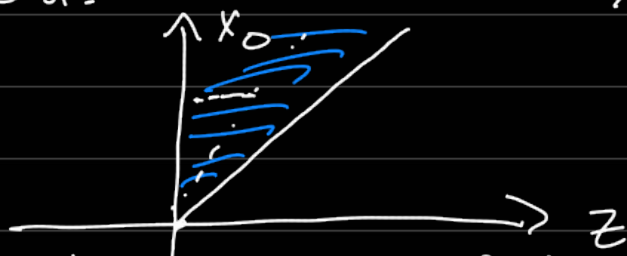
$$\Rightarrow \frac{dz^2}{z^2} = \frac{du^2}{u^4}$$

$$ds^2 = \frac{L^2}{z^2} (-dz^2 + \sum_{i=1}^{D-2} dx^i dx^i)$$

$$D-2$$

$$ds^2 = \frac{1}{z^2} (dz^2 - (dx^0)^2 + \sum_{i=1}^n (dx^i)^2) \quad (X^0, X^1, \dots, X^n)$$

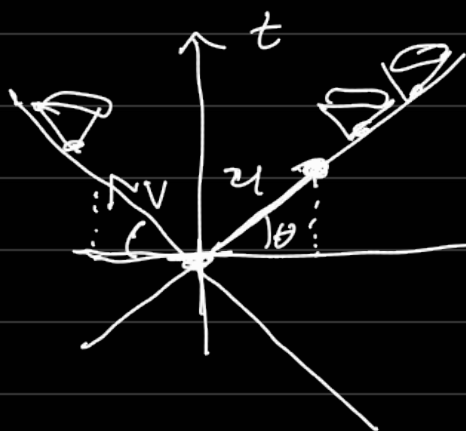
- Which is conformally eq. to MD with $z > 0$
 $\Rightarrow X^i = \text{const.} \quad X_0 = z$



$z \rightarrow 0$: Reach the Boundary of the 'poincare' patch which is the conformal compactification of MD

ex: $M^{1,1}$

$$ds^2 = -dt^2 + dx^2, \quad -\infty < t, x < \infty$$



$$\tan(\theta) = \frac{t}{x}$$

$$\begin{cases} u = x + t \\ v = t - x \end{cases}$$

$$t = \frac{1}{2}(u + v)$$

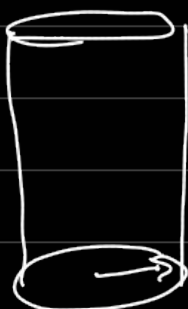
$$x = \frac{1}{2}(u - v)$$

$$dt^2 = \frac{1}{4}(du^2 + 2du dv + dv^2)$$

$$dx^2 = \frac{1}{4}(du^2 - 2du dv + dv^2)$$

$$ds^2 = -du dv$$

* Now map to a cylinder

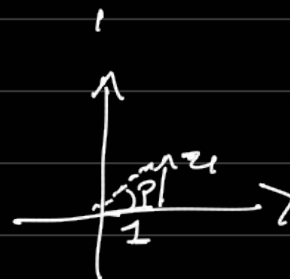


$$u = \tan(P)$$

$$v = \tan(Q)$$

$$-\infty : \cos(-\pi/2)$$

$$+\infty : \cos(\pi/2)$$



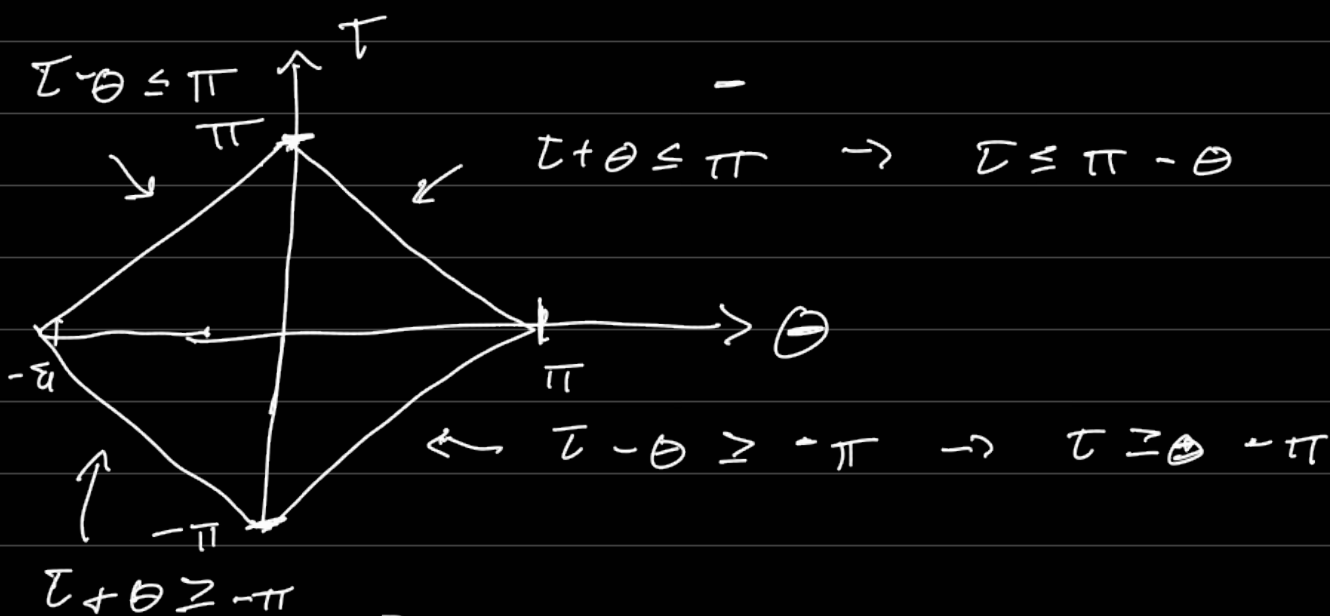
$$P, Q \in (-\pi/2, \pi/2)$$

$$T \equiv Q - P \in (-\pi, \pi)$$

$$\Theta \equiv Q + P \in [0, \pi)$$

$$ds^2 = \frac{1}{4 \cos^2(\Theta) \cos^2(T)} (-dT^2 + d\Theta^2)$$

$$|\phi \pm p| \leq \pi$$



* Now for M^3 : $ds^2 = -dt^2 + dr^2 + r^2 d\theta^2$

* $u = t + r$ | $r = \frac{u - v}{2}$ $\tau \geq 0$
 $v = t - r$ $-\infty < t < \infty$

$u = \tan(P)$ | $ds^2 = -du dv + \left(\frac{u-v}{2}\right)^2 d\theta^2$
 $v = \tan(Q)$ $u > v$ $P > Q$

$T \equiv Q + P \Rightarrow Q = (T+R)/2 : |Q| \leq \frac{\pi}{2}$
 $R = Q - P$ $P = (T-R)/2 : |P| \leq \frac{\pi}{2}$
 $P > Q \Rightarrow T+R > T-R \Rightarrow R > 0$

* If $f(x) > f(y)$ then $x > y$

with f bijective on its domain

$$f(x) - f(y) > 0$$

Since f is bijective and continuous

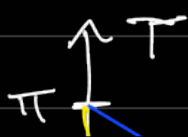
$$f'(x) \approx \frac{f(x) - f(y)}{x - y}$$

$$f(x) - f(y) \approx f'(x)(x - y)$$

$$\Rightarrow f(x) - f(y) > 0 \Rightarrow f'(x)(x - y) > 0$$

$$\tan(x) = \sec^2(x) \Rightarrow f'(x) > 0$$

$$\Rightarrow (x - y) > 0 \quad \square$$



$$\left| \frac{T+R}{2} \right| \leq \frac{\pi}{2}$$

$$\frac{|T-R|}{2} \leq \frac{\pi}{2}$$



$$T+R \leq \pi$$

$$T+R \geq -\pi$$

$$\hookrightarrow T \geq -R-\pi$$

does not exist
for $R > 0$

$$T-R \leq \pi$$

$$T-R \geq -\pi$$

i^0 : Spatial infinity

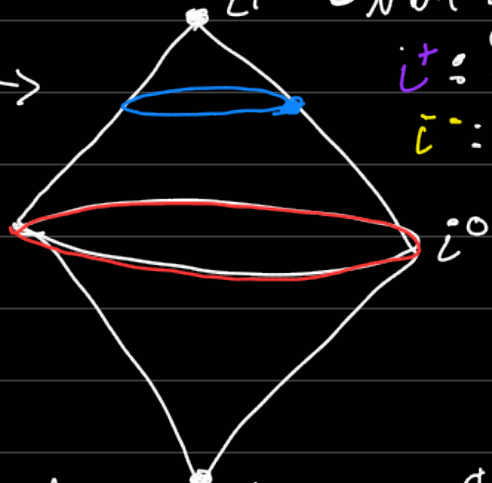
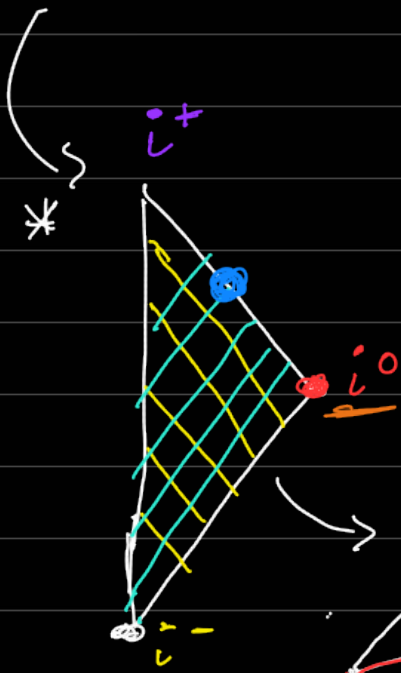
• Each point has an S^2 (in M^4)

so if we expand $\theta \in [0, \pi]$

if • Null geodesics are

i^+ : future time-like infinity

i^- : past time-like infinity



$$\tilde{d}_{S^2}^2 = (-dT^2 + dR^2 + \sin^2(\chi)(d\theta^2 + \sin^2(\theta)dp^2))$$

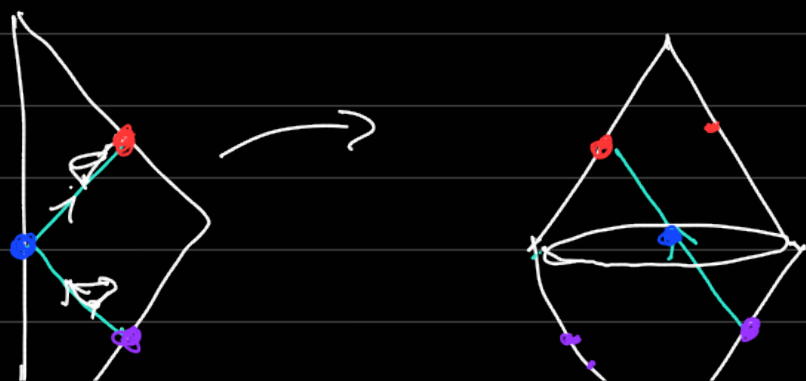
$\Rightarrow$ Null geodesics: $dT = \pm dR$

$$\Rightarrow T = \pm R + C$$

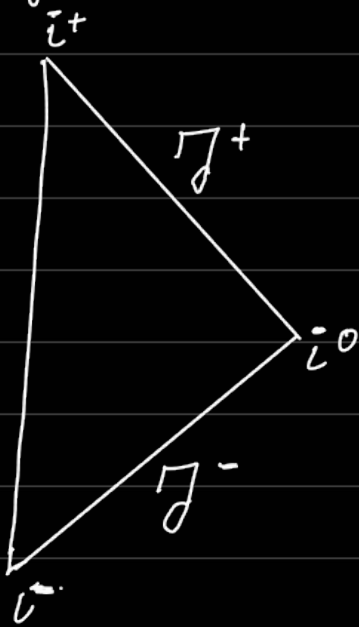
$$T+R = C$$

$$T-R = C$$

*The "Bouncing of Boundary" I saw earlier was simply the Penrose diagram without the S^2 element



In general:



i^0 : Space-like infinity
 i^+ : future timelike inf.
 i^- : Past timelike inf.
 J^- : Past Null inf.
 J^+ : Future Null inf.

(i) Space-like geodesics start and end at i^0

(ii) Null-geodesics

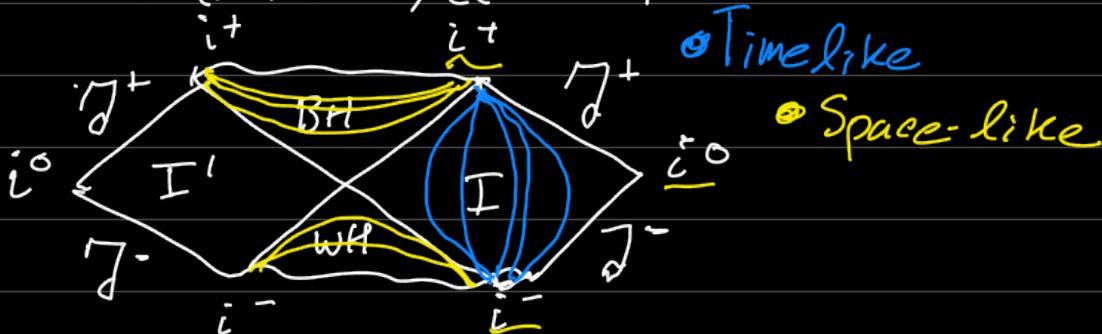
- Start on J^-
- End on J^+

(iii) Time-like:

- Start at i^-
- End at i^+

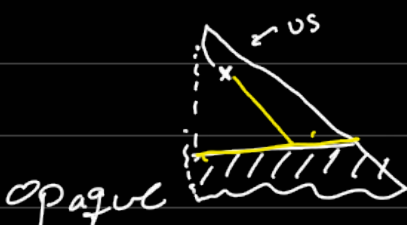
* Other space times:

(a) Schd (maximally extended)

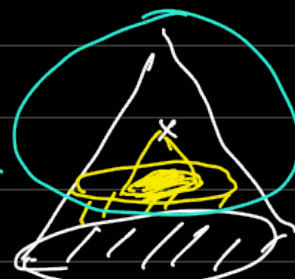


(b) Isotropic/homogeneous universe:

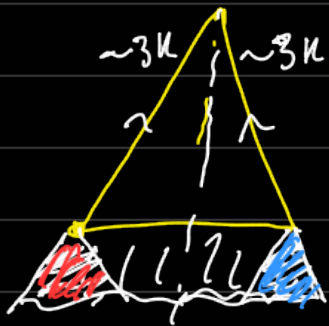
$$K=0, \Lambda=0$$



S^1



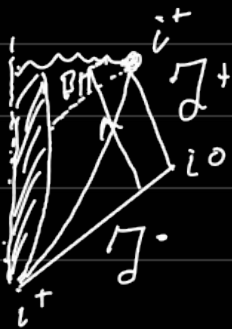
→ Big Bang



• Which ever process occurred to bring the photon on the R (via ~~light~~) and L (via ~~light~~) could not have communicated.

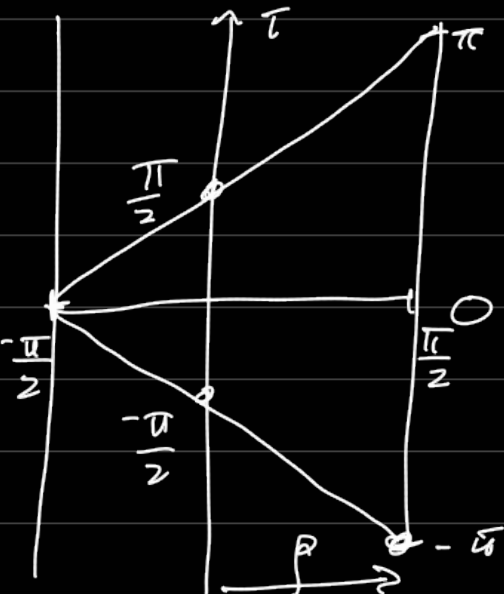
Cosmological Horizon problem
↳ Inflation

© Gravitational Collapse:



Back to AdS: with

$$ds^2 = \frac{L^2}{z^2} \left(dz^2 - (dx^0)^2 + \sum_i (dx^i)^2 \right)$$



(ρ, τ) : Global coords (G)
 (z, x^0) : local Poincaré coords (P)

* $\tau^0 = L \sqrt{1 + r^2/L^2} \sin(\frac{\tau}{L})$, where $\sqrt{1 + r^2/L^2} = \frac{1}{\cos(\rho)}$
 $= L \sin(\tau)/\cos(\rho)$ and $\tau = t/L$

$r^2 = r$
 $\hookrightarrow \left(\frac{1 - c^2}{c^2} \right) L^2 = r^2 \Rightarrow \underline{r = L \frac{\sin(\rho)}{\cos(\rho)}}$

$c^2 = 1 - r^2/L^2$

$$= \frac{L \cos(\tau)}{\cos(\rho)} \quad (\text{Global AdS}_2)$$

=> Comparing to Poincare to get the 2D coords:

	(Global)	(Poincare)
y_0	$\frac{L \sin(\tau)}{\cos(\rho)} = L z t$	
y^1	$L \frac{\sin(\rho)}{\cos(\rho)} = \left(\frac{1}{2L}\right) (-1 + z^2 (L^2 + t^2))$	
y^2	$L \frac{\cos(\tau)}{\cos(\rho)} = \frac{1}{2L} (1 + z^2 (L^2 - t^2))$	

solve for (t, z) in terms of (τ, ρ)

i) $\frac{\sin(\tau)}{\cos(\rho)} = t \frac{1}{z}$

ii) $L \frac{\sin(\rho)}{\cos(\rho)} = \frac{z}{2} \left(-1 + \frac{1}{z^2} (L^2 + t^2) \right)$

iii) $L \frac{\cos(\tau)}{\cos(\rho)} = \frac{z}{2} \left(1 + \frac{1}{z^2} (L^2 - t^2) \right)$

$$L \frac{(\sin(\rho) + \cos(\tau))}{\cos(\rho)} = \frac{z}{2} \left(\frac{1}{z^2} L^2 \right)$$

$$z = \frac{L \cos(\rho)}{\sin(\rho) + \cos(\tau)}$$

and so $t = \frac{\sin(\tau)}{\cos(\rho)} z = \frac{L \sin(\tau)}{\sin(\rho) + \cos(\tau)}$

=> $t = \frac{L \sin(\tau)}{\cos(\tau) + \sin(\rho)}$, $z = \frac{L \cos(\rho)}{\cos(\tau) + \sin(\rho)}$

* Observers that follow curves of constant z reach the Boundary point: $(\tau = \pi, \rho = \pi/2)$ as $t \rightarrow \infty$

$z = \infty$: "Horizon"

$$\cos(\tau) + \sin(\rho) = 0$$

=> Restricted to: $\sin\left(\tau + \frac{\pi}{2}\right) = \cos(\tau)$

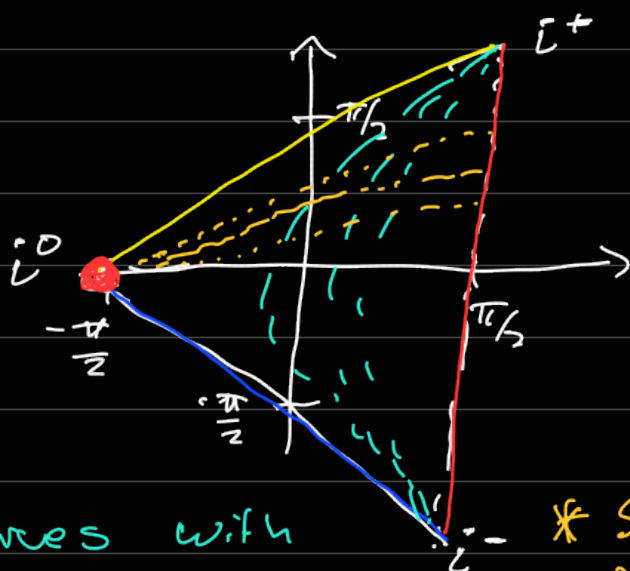
=> $\sin\left(\tau + \frac{\pi}{2}\right) + \sin(\rho) = 0$

$\rho = -\left(\tau + \frac{\pi}{2}\right) \Rightarrow \tau = -\rho - \pi/2$
 (a) $\tau = \rho + \frac{\pi}{2}$
 (b) $\tau = -\rho - \pi/2$

$-\sin\left(\tau - \frac{\pi}{2}\right) + \sin(\rho) = 0 \Rightarrow$ (a) $\tau - \frac{\pi}{2} = \rho \Rightarrow \tau = \rho + \frac{\pi}{2}$

Thus: Horizon gives

$$\underline{\tau = \rho + \pi/2} \wedge \underline{\tau = -\rho - \pi/2}, \text{ since } \rho \in [-\pi/2, \pi/2]$$



* Time-like curves with fixed z

* Space-like curves fixed t

Boundary of patch is the (disconnected) union of $\{0 < t < \infty \text{ at } \rho = \pi/2\}$ and the single point $(t=0 \text{ at } \rho = -\pi/2)$

* Geodesics: Curve fixed at the spatial origin
 $\tau = 0 = L \frac{\sin(\rho)}{\cos(\rho)} \Rightarrow \rho = 0$

So any point can be brought to the origin

Isometries of AdS_2 $(x^0, x^i) = (D-1)$ coords x^0
 i.e. $x^0 \in M^{D-1}$

(i) Translations $(D-1)$ x^0

(ii) $(D-1)(D-2)/2$ Lorentz of x^0

(iii) Dilation/scale $\underline{I}$: $\delta x^0 = \rho x^0, \delta x^i = \rho x^i, \delta z = \rho z$

(iv) $(D-1)$ SCT: $\delta x^0 = C^0 (x \cdot x + z^2) - 2C \cdot x x^0$
 $\delta z = -2C \cdot x z$

$$C^\mu = (C^0, C^z = 0)$$

$\Rightarrow D(D+1)/2$: transformations/generators: $SO(D-1, 2)$

* Patch and Global have the same isometry group but each act differently

$\Rightarrow$ LAB can cross the Horizon

* In the limit $z \rightarrow 0$ we get the normal SCT

→ Isometry group of the Poincaré Patch of AdS_D
acts as the conformal group of its M^{D-1} Boundary

Claim: the above are Killing. $|ds^2 = \frac{L^2}{z^2} (dz^2 - (dx^0)^2 + \sum_i (dx^i)^2)$

$$x'^\mu = x^\mu + \epsilon K^\mu(x)$$

$$\Rightarrow \mathcal{L}_K g = \nabla_\mu K_\nu + \nabla_\nu K_\mu = K^\sigma \partial_\sigma g_{\mu\nu} + g_{\mu\sigma} \partial_\mu K^\sigma + g_{\nu\sigma} \partial_\nu K^\sigma$$

$$\leadsto M = (\sigma, z) \quad \sigma = 0, \dots, D-1, \quad D = z$$

Break down into (μ, ν) pairs

(μ, ν) :

$$\begin{aligned} (z, z): \quad K^\sigma \partial_\sigma \left(\frac{L^2}{z^2} \right) + 2 g_{z\sigma} \partial_z K^\sigma &= K^z \left(-\frac{2L^2}{z^3} \right) + 2 \left(\frac{L^2}{z^2} \right) \partial_z K^z \\ &= \left(-K^z \frac{2L^2}{z^3} \right) + 2 \left(\frac{L^2}{z^2} \right) \partial_z K^z = 0 \\ &= K^z - z \partial_z K^z = 0 \end{aligned}$$

$$\begin{aligned} (0, 0): \quad K^z \left(+\frac{2L^2}{z^3} \right) - 2 g_{00} \partial_0 K^0 &= 0 \\ K^z - z \partial_0 K^0 &= 0 \end{aligned}$$

$$\begin{aligned} (i, i): \quad \left(-\frac{L^2}{z^3} \right) K^z + g_{ii} \partial_j K^i + g_{jj} \partial_i K^j &= 0 \\ i \neq j \quad (\partial_i K^j + \partial_j K^i) &= 0 \end{aligned}$$

$$(i, i): \quad K^z - z \partial_i K^i = 0$$

$$\begin{aligned} (z, 0): \quad g_{zz} \partial_0 K^z + g_{00} \partial_z K^0 &= 0 \\ \frac{L^2}{z^2} \partial_0 K^z - \frac{L^2}{z^2} \partial_z K^0 &= 0 \\ \partial_0 K^z - \partial_z K^0 &= 0 \end{aligned}$$

$$(z, i): \quad \partial_i K^z + \partial_z K^i = 0$$

$$(i, 0): \quad \partial_0 K^i - \partial_i K^0 = 0$$

• From $(z, z): \quad K^z - z \partial_z K^z = 0$

$$K^z = f(x) z \Rightarrow f(x) z - z f(x) = 0 \quad \checkmark$$

$$\underline{K^z(x, z) = f(x) z}$$

$$(0, 0): \quad f(x) z - z \partial_0 K^0 = 0$$

$$\Rightarrow \begin{cases} \partial_0 K^0 = f(x) \\ \partial_i K^i = f(x) \\ \partial_z K^0 = z \partial_0 f(x) \\ \partial_z K^i = -z \partial_i f(x) \\ \partial_0 K^i - \partial_i K^0 = 0 \end{cases}$$

$$\partial_i k^i + \partial_i k^i = 0$$

$$(a) : f(x) = 0 \Rightarrow k^0 = a^0, k^i = a^i$$

and so $K = a^\mu \partial_\mu$ which are translations

: Also have $K = x^i \partial_i - x^0 \partial_0$; $K = x^i \partial_i + x^0 \partial_0$

Lorentz

$$(c) f(x) = \rho : K = \rho x^\mu \partial_\mu \text{ (scale)}$$

$$(d) \text{ Assume } k^i = 0$$

$$\partial_z k^0 = z \partial_z^2 k^0$$

$$(k^0) = (x^0)^2 + z^2 + g(x^i), \quad g(x^i) = 0$$

$$\Rightarrow f(x) = 2x^0$$

$$K = ((x^0)^2 + z^2) \partial_0 + 2x^0 z \partial_z$$

* The Killing vector for the scale transformation above agrees with $L_{D(1)}$

$$K = \rho D = \rho x^\mu \partial_\mu, \quad \mu = 0 = z$$

$\Rightarrow$ Transform from $(x^0, z) \rightarrow y^A$

$$L_{D(1)} = y_0 \frac{\partial}{\partial y^0} - y_{0,z} \frac{\partial}{\partial y^0}$$

